

Video Displays

Display is a device used to present visual information. The main aim of any display technology is to simplify the information sharing. Video Displays are two-dimensional displays that cover a full area, basically a rectangle. The different types of video displays are discussed below

Electroluminescent Display

Electroluminescent display is a flat panel display that works on the principle of electroluminescence. Electroluminescence is a phenomenon where materials can emit light in response to the electric current.

Advantages

- Thin and compact.
- Good writing speed.
- Low voltage operation.
- Disadvantages
- Low efficiency.
- High cost.
- High voltage drivers.

Electronic Paper

Electronic Paper also known as e-paper is a term for display technologies resembling a paper. The popularity of the display is due to its contrast, thickness, readability and flexibility.

E Ink is now one of the popular e-paper display technologies. It was developed by E Ink Corporation and today they are used by different companies in their products and e-readers. E Ink displays are made of some tiny capsules which consist of black and white particles.

Advantages

Less power.

Easy to read at outdoors.

Disadvantages

Can't display full videos.

Cannot be read in dark.

LED Display

It is a type of display that uses light emitting diodes. An LED display consists of many closely-spaced LEDs. By varying the brightness of each LED, the diodes jointly form an image on the display.

To create a bright colour image, the principles of additive colour mixing are used, whereby new colours are created by mixing light in different colours.

ADVANTAGES:

Long Service Life
Good environmental performance
Low heat generation
Low Power

DISADVANTAGES:

Sensitive to Voltage Spike
Heat dissipation in some applications

Liquid Crystal Display(LCD)

This display generally depends upon the light modulating properties of liquid crystals. The low power consumption makes it good to use in all types of electronic equipments that are battery powered.

LCDs are commonly used in systems, such as calculators and laptop computers.

Plasma Display

Plasma panels, also called gas discharge displays, are constructed by filling the region between two glass plates with a mixture of gases that usually include neon. A series of vertical conducting ribbons is placed on one glass panel, and a set of horizontal ribbons is built into the other glass panel. Xenon and Neon are most commonly used gases to make plasma.

OLED DISPLAY

OLED means Organic Light Emitting Diode. As the name indicates that it relies on organic materials. Organic Light Emitting Devices (OLED) emit light from active luminescent material in each display pixel.

AMOLED DISPLAY

The full form of AMOLED is Active Matrix OLED. It has better display quality, thin form factor, and lower power consumption. In this display technology a very thin film has been used which was coated with several organic electroluminescent compounds.

Analog and DVI Connections

To display information on a monitor, your computer sends the monitor a signal. The signal can be in analog or digital format.

Analog (VGA) Connection

Because most CRT monitors require the signal information in analog (continuous electrical signals or waves) form and not digital (pulses equivalent to the binary digits 0 and 1), they typically use an analog connection.

However, computers work in a digital world. The computer and video adapter convert digital data into analog format. A video adapter is an expansion card or component that provides the ability to convert display information into a signal that is sent to the monitor. It can also be called a graphics adapter, video card or graphics card.

DVI Connection

DVI keeps data in digital form from the computer to the monitor. There's no need to convert data from digital information to analog information. LCD monitors work in a digital mode and support the DVI format. (Although, some also accept analog information, which is then converted to digital format.) At one time, a digital signal offered better image quality compared to analog technology. However, analog signal processing technology has improved over the years and the difference in quality is now minimal.

DVI signals for CRT monitor

The incorporation of DVI with a traditional analogue CRT monitor will allow monitors to be designed to receive digital signals, with the necessary digital-to-analogue conversion being carried out within the monitor itself. This will give manufacturers added control over final image quality, making differentiation based on image quality much more of a factor than it has been hitherto. However, the application of DVI with CRT monitors is not all plain sailing. With a selective refresh interface, DVI can maintain the high refresh rates required to keep a CRT display ergonomically pleasing while avoiding an artificially high data rate between the graphics controller and the display.

There are two main types of DVI connections:

DVI-digital (DVI-D) is a digital-only format. It requires a video adapter with a DVI-D connection and a monitor with a DVI-D input. The connector contains 24 pins/receptacles in 3 rows of 8 plus a grounding slot for dual-link support. For single-link support, the connector contains 18 pins/receptacles.

DVI-integrated (DVI-I) supports both digital and analog transmissions. This gives you the option to connect a monitor that accepts digital input or analog input. In addition to the pins/receptacles found on the DVI-D connector for digital support, a DVI-I connector has 4 additional pins/receptacles to carry an analog signal.

LCD Panels

LCD panels have now largely supplanted CRT displays in new computer installations. That isn't to say CRTs are obsolete far from it. They are now just as common. Desktop LCD panels use technology similar to that of LCDs in laptop computers. Compared to the original display technology (CRTs), LCDs have low-glare, completely flat screens and low power requirements (5 watts versus nearly 100 watts for an ordinary CRT monitor). The color quality of a good active-matrix LCD panel actually exceeds that of most CRT displays, particularly when viewed from head on.

Video Display Adapters

A video display adapter (aka video card) provides the interface between your computer and your monitor and transmits the signals that appear as images on the display.

The following list of standards can serve as an abbreviated history of PC video-display technology:

- MDA (Monochrome Display Adapter)
- HGC (Hercules Graphics Card)
- CGA (Color Graphics Adapter)
- EGA (Enhanced Graphics Adapter)
- VGA (Video Graphics Array)
- SVGA (Super VGA)
- XGA (Extended Graphics Array) and beyond

Video Adapter Types

A monitor requires a source of input. The signals that run to your monitor come from one or more

video display adapters in the system.

There are four basic types of video display adapters:

■ **Discrete plug-in video cards**—These cards require the use of an expansion slot but provide the highest possible level of features and performance.

■ **Discrete video on the motherboard**—The same discrete circuitry that can be found on a video card can also be directly built in or mounted on the motherboard. This is how high-end video is installed in modern laptops and some older desktop systems; however, modern desktops normally use either discrete video on a plug-in card or motherboard chipset or processor integrated video.

■ **Motherboard chipset integrated video**—This is video circuitry integrated into the motherboard chipset. Because it shares the system random access memory (RAM) and other components, it provides an economical solution that also uses less power (ideal for laptops).

■ **Processor integrated video**—This is video circuitry integrated into the processor, either as a separate die within the processor package or directly in the processor die. This form of integrated

video also shares the system RAM and requires a motherboard with specific chipsets and video interface connectors on-board. Because chipset or processor-based integrated video shares the system RAM and other components, it has the lowest cost of any video solution. Performance is generally lower than that of a separate card-based adapter, especially for 3D gaming or other graphics-intensive applications. Resoluti

Video Adapter Components

All video display adapters contain certain basic components, such as the following:

- The video BIOS
- The video processor
- The video memory
- The digital-to-analog converter (DAC)
- The bus connector
- The video driver

Many of the popular adapters on the market today include additional modules intended

for special purposes, such as 3D acceleration. The following sections examine these components in greater detail.

The Video BIOS. Video adapters include a BIOS (basic input/output system) that is similar in construction, but completely separate from the main system BIOS. (Other devices

in your system, such as SCSI adapters, may also include their own BIOS.) If you turn your monitor on first and look quickly, you may see an identification banner for your adapter's video BIOS at the very beginning of the system startup process.

The Video Processor

The video processor, or chipset, is the heart of any video adapter and essentially defines the card's functions and performance levels. Two video adapters

built using the same chipset often have many of the same capabilities and deliver comparable performance. Also, the software drivers that operating systems and applications

use to address the video adapter hardware are written primarily with the chipset in mind. You can often use a driver intended for an adapter with a particular chipset on any other adapter using the same chipset. Of course, cards built using the same chipset can differ in the amount and type of memory installed, so performance may vary.

There are three main types of processors used in video adapters:

- Frame buffers
- Coprocessors
- Accelerators

The oldest technology used in creating a video adapter is known as frame-buffer technology. In this scheme, the video adapter is responsible for displaying the individual frames of an image.

The Video RAM.

Video adapters rely on their own on-board memory that they use to

store video images while processing them. The amount of memory on the adapter determines the maximum screen resolution and color depth that the device can support. You

can often select how much memory you want on a particular video adapter—for example, 256K, 512K, 1M, 2M, 4M, or 8M are common choices today. Most cards today

come with at least 2M, and many have 4M. Adding more memory does not speed up your video adapter; instead, it enables the card to generate more colors and/or higher resolutions.

RAM Calculations.

The amount of memory that a video adapter needs to display a particular resolution and color depth is based on a mathematical equation. There has to be a location in the adapter's memory array to display every pixel on the screen, and the number of total pixels is determined by the resolution. For example, a screen resolution of 1,024×768 requires a total of 786,432 pixels.

Here

is how the calculation works:

$$\begin{aligned} 1,024 \times 768 &= 786,432 \text{ pixels} \times 4\text{-bits per pixel} \\ &= 3,145,728 \text{ bits} \\ &= 393,216 \text{ bytes} \\ &= 384\text{K} \end{aligned}$$

Video Bus Width. Another issue with respect to the memory on the video adapter is the width of the bus connecting the graphics chipset and the memory on the adapter. The chipset is usually a single large chip on the card that contains virtually all of the adapter's functions. It is wired directly to the memory on the adapter through a local bus on the card.

The Digital-to-Analog Converter. The digital-to-analog converter on a video adapter

(commonly called a RAMDAC) does exactly as its name describes. The RAMDAC is responsible for converting the digital images that your computer generates into analog

signals that the monitor can display. The speed of the RAMDAC is measured in MHz; the faster the conversion process, the higher the adapter's vertical refresh rate. The speeds of the RAMDAC's used in today's high performance video adapters can exceed 200MHz.

The Video Driver.

The software driver is an essential, and often a problematic, element of a video display subsystem. The driver enables your software to communicate with the video adapter. You can have a video adapter with the fastest processor and the most efficient memory on the market, but still have poor video performance because of a badly written driver.

DOS applications address the video display hardware directly, and typically include their own drivers for various types of video adapters. All versions of Windows, however, use a

driver that is installed in the operating system. Applications then can use operating system function calls to access the video hardware.

Video Cards for Multimedia

Multimedia has become an important element of the personal computing industry.

Technology that was at first relegated to entertainment uses, and therefore dismissed by many computing professionals, has now been assimilated into the corporate world. Live

data feeds, videoconferencing, and animated presentations are just a few of the technologies that are now commonplace elements of business computing.

Video Output Devices. When video technology was first introduced, it was based on television. There is a difference between the signals used by a television and the signals used by a computer. In the United States, the National Television System Committee (NTSC) established color TV standards in 1953. Some other countries, such as Japan,

followed this standard. Many countries in Europe developed more sophisticated standards, including Phase Alternate Line (PAL) and SEquential Couleur Avec Mémoire

(SECAM). Table 8.9 shows the differences among these standards.

Still-Image Video Capture Cards. You can install a board into your PC that enables you to capture individual screen images for later editing and playback. There are also external devices that plug into a PC's parallel port. These units capture still images from NTSC video sources such as camcorders or VCRs. Although image quality is limited by

the input signal, the results are still good enough for presentations and desktop publish-

ing applications. These devices work with 8-, 16-, and 24-bit VGA cards and usually accept video input from VHS, Super VHS, and Hi-8 devices. As you might expect, however,

Super VHS and Hi-8 video sources give better results, as do configurations using more than 256 colors.

Multiple Monitors. Windows 98 includes a video display feature that Macintosh systems have had for years: the capability to use multiple monitors on one system. Windows 98 supports up to nine monitors (and video adapters), each of which can provide a

different view of the desktop. When you configure a Windows 98 system to do this, the operating system creates a virtual desktop; that is, a display that exists in video memory that can be larger than the image actually displayed on a single monitor. You use the multiple monitors to display various portions of the virtual desktop, enabling you to place the windows for different applications on separate monitors, and move them around at will.

Desktop Video (DTV) Boards. You can also capture NTSC (television) signals to your computer system for display or editing. In other words, you can literally watch TV in a window on your computer. When capturing video, you should think in terms of digital

versus analog. The biggest convenience of an analog TV signal is efficiency; it is a compact way to transmit video information through a low-bandwidth pipeline. The disadvantage is that while you can control how the video is displayed, you can't edit it.

Actually capturing and recording video from external sources and saving the files onto your PC requires special technology. To do this, you need a device called a video capture board, also called a TV tuner, video digitizer, or video grabber.

Codec:

result. There are two types of codecs: hardware-dependent codecs and software (or hardware-independent) codecs. Hardware codecs typically perform better; however, they require additional hardware. Software codes do not require hardware for compression or playback, but they typically do not deliver the same quality or compression ratio. Two of the major codec algorithms are:

■ **JPEG (Joint Photographic Experts Group).** Originally developed for still images, JPEG can compress and decompress at rates acceptable for nearly full-motion video (30fps). JPEG still uses a series of still images, which makes editing easier. JPEG is

typically lossy (meaning that a small amount of the data is lost during the com-

pression process, slightly diminishing the quality of the image), but it can also be

lossless. JPEG compression functions by eliminating redundant data for each individual image (intraframe). Compression efficiency is approximately 30:1 (20:1–

40:1).

■ **MPEG (Motion Pictures Experts Group).** MPEG by itself compresses video at approximately a 30:1 ratio, but with pre-compression through oversampling, the ratio can

climb to 100:1 and higher, while retaining high quality. Thus, MPEG compression results in better, faster videos that require less storage space. MPEG is an interframe compressor. Because MPEG stores only incremental changes, it is not used during editing phases. MPEG decoding can be performed in software on high-performance Pentium systems.

3D Graphics Accelerators

The latest trend in PC graphic display technology is the expanded use of three dimensional images. Three-dimensional—or 3D—imagery has been used for years in the computer gaming world, and has even made its way into business computing. Spreadsheet programs such as Microsoft Excel have for years used 3D charts to display spreadsheet data. Of course, the images are not truly three-dimensional, because the monitor screen is a 2-D medium, but software can produce 3D effects using perspective, textures, and shading.

The basic function of 3D software is to convert image abstractions into the fully realized images that are then displayed on the monitor. The image abstractions typically consist of the following elements:

■ **Vertices.** Locations of objects in three dimensional space, described in terms of their X, Y, and Z coordinates on three axes representing height, width, and depth.

■ **Primitives.** The simple geometric objects that the application uses to create more complex constructions, described in terms of the relative locations of their vertices. This serves not only to specify the location of the object in the 2D image, but

also provides perspective, as the three axes can define any location in three-dimensional space.

■ **Textures.** Two-dimensional bitmap images or surfaces that are designed to be mapped onto primitives. The software enhances the 3D effect by modifying the appearance of the textures depending on the location and attitude of the primitive; this process is

called perspective correction. Some applications use another process called MIP mapping, which utilizes different versions of the same texture that contain varying

amounts of detail, depending on how close the object is to the viewer in the three-dimensional space. Another technique, called depth cueing, reduces the color and

intensity of an object's fill as the object moves farther away from the viewer.

Rendering

Rendering depends on two standardized functions that convert the abstractions into the completed image that is displayed onscreen.

The standard functions performed in rendering are:

■ **Geometry.** The sizing, orienting, and moving of primitives in space and the calculation of the effects produced by the virtual light sources that illuminate the image.

■ **Rasterization.** The converting of primitives into pixels on the video display by filling the shapes with properly illuminated shading, textures, or a combination of the two.

Rasterization

A modern video adapter that includes a chipset capable of 3D video acceleration will have special built-in hardware that can perform the rasterization process much faster than if it is done by software (using the system processor) alone. Most chipsets with 3D acceleration perform the following rasterization functions right on the adapter: 545 p6v6 Upgrading & Repairing 1636-4
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■ **Scan conversion.** The determination of which onscreen pixels fall into the space delineated by each primitive.

■ **Shading.** The process of filling pixels with smoothly flowing color using the flat or Gouraud shading technique.

■ **Texturing.** The process of filling pixels with images derived from a 2D sample picture or surface image.

■ **Visible surface determination.** The identification of which pixels in a scene are obscured by other objects closer to the viewer in three-dimensional space.

■ **Animation.** The process of switching rapidly and cleanly to successive frames of motion sequences.

3D Chipsets. As with standard 2D video adapters, there are several manufacturers of popular 3D video chipsets, and many more manufacturers of video adapters that use them. Table 8.10 lists the major 3D chipset manufacturers, the various chipsets they make, and the video adapters that use them.